Subhash Rana

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Summary

Motivated Computer Science undergraduate with practical internship experience in full-stack MERN development and solid foundations in Data Structures & Algorithms. Adept at engineering scalable web applications that deliver measurable performance gains. Eager to contribute to collaborative teams tackling impactful, real-world software challenges.

Technical Skills

Languages: C++, JavaScript, TypeScript, Python
AI / LLM: Gemini, LLMs, AI Agents, Vercel AI SDK, Tool Calling, Prompt Engineering, Zod
Frontend: HTML5, CSS3, Tailwind CSS, React.js, Next.js, Ink
Backend: Node.js, Express.js, REST APIs
Databases: MongoDB, PostgreSQL, SQL, Prisma ORM
Developer Tools: Git, GitHub, Postman, VS Code
Coursework: Data Structures & Algorithms, DBMS, Operating Systems

Experience

Software Development Intern Dec 2025 – Mar 2026
Matchbest Software Pvt. Ltd. *Haryana, India*

- **EMS Portal** – Engineered key modules of an Employee Management System using Next.js, targeting two core workflows: Leave Management and Reimbursement Management.
- Architected the Leave Application module, empowering employees to submit, track, and monitor leave requests with real-time status updates and admin approval workflows.
- Implemented the Reimbursement Application module, streamlining expense claim submissions with document uploads and equipping finance teams with a structured review and approval pipeline.
- Crafted RESTful APIs and enforced role-based access control (RBAC) to safeguard data across employee and admin roles.
- **Mavericksservices** – Constructed responsive UI components using Next.js and Tailwind CSS, guaranteeing cross-browser compatibility and seamless mobile responsiveness.
- **Ava CRM** – Spearheaded frontend and backend development of interactive dashboards, lead tracking modules, and client data features; leveraged PostgreSQL with Prisma ORM for optimized data retrieval and storage.

Projects

Job Portal | *React.js, Node.js, Express.js, MongoDB, REST APIs*

- Delivered a full-stack MERN job portal featuring job listings, candidate applications, and recruiter workflows.
- Integrated JWT-based authentication, role-based access control, and an admin dashboard to oversee postings.
- Boosted API performance via query indexing and pagination, cutting average response time by 25% and boosting posting efficiency by 40%.

Pilot CLI | *TypeScript, Node.js, Vercel AI SDK, OpenAI, React, Ink, Zod*

- Built a terminal-based AI coding agent capable of understanding user requests, selecting tools, executing multi-step workflows, and generating contextual responses.
- Implemented an agent loop with LLM tool calling, supporting file operations, shell execution, web search, and code execution with tool-result feedback.
- Added human-in-the-loop approval workflows for sensitive tool operations and used Zod for runtime validation of tool inputs before execution.
- Developed agent evaluations using mocked tools to measure tool selection, tool ordering, forbidden-tool avoidance, and multi-turn task performance.

Jira-Style Task Management System | *React, Node.js, Express, MongoDB, TanStack Query, Docker*

- Engineered a full-stack task management platform featuring an interactive Kanban board, drag-and-drop workflows, and optimistic UI updates using TanStack Query.
- Implemented JWT authentication with HTTP-only cookies, bcrypt password hashing, Zod validation, and ownership-based authorization to prevent IDOR vulnerabilities.
- Developed secure file attachments with S3-compatible storage and binary file validation; added 308 automated frontend and backend tests.

Education

Bachelor of Technology in Computer Science Engineering 2022 – 2026
St. Andrews Institute of Technology and Management | *CGPA:* ***7.75*** *Haryana, India*

Senior Secondary – CBSE (Science) | **80.4%** 2020 – 2021
Cambridge Public School *Hisar, Haryana*

Secondary – HBSE | **84.4%** 2018 – 2019
St. Paul High School *Hisar, Haryana*